ACCORDLY

getting you that sweet reward of a sweet new perfume

Vala

WHY

The Culprit: Commodity Paper-

Sold as a “personal” and “skin-close” scent just for me
Instead, it was so strong I was trying to run away from myself and ended up with a migraine I couldn't scrub off :(

on paper, it looked right

- ★ sandalwood, amber, cedarwood
  - ☆ notes I already know I like
- ★ Iso E Super
  - ☆ wrongly compared to Ambroxan
- ★ the logic was there
  - ☆ but it still failed!

this isn’t just a one-off

- ★ rating-based tools prioritise popularity
  - ☆ not composition
- ★ composition tools (Scent IQ, PERFUMIST) do exist
  - ☆ but they’re still “black boxes”
- ★ nothing is transparent
  - ☆ LET ME SEE >:(

WHAT I BUILT

A fragrance recommender

made for those** (me) **who want** (need) **to know

MOST TOOLS

my input

→ opaque calculation

→ mystery result

MINE

my input

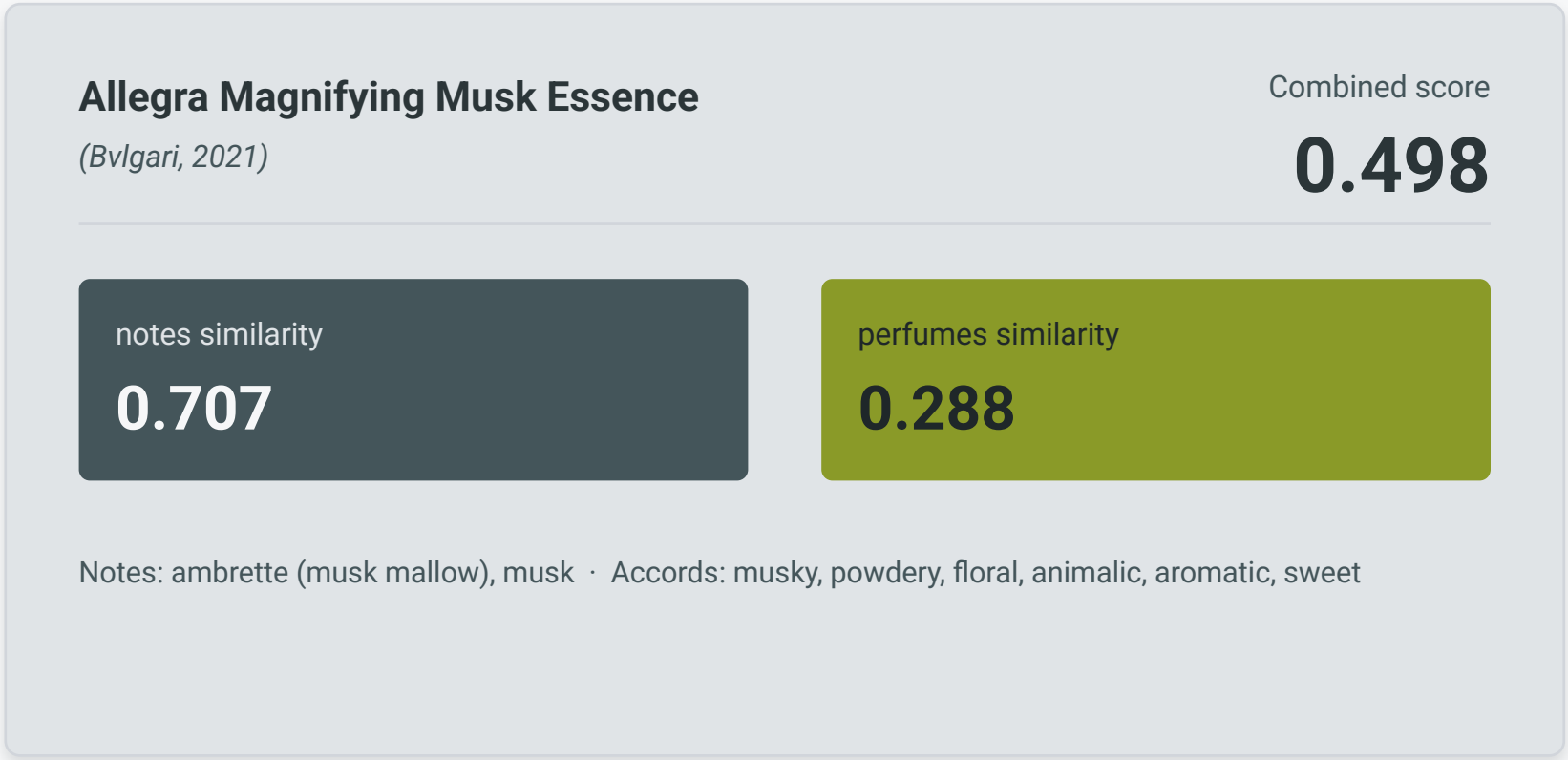
→ transparent breakdown

→ now I know :D

WHAT I BUILT

Transparent by design

every recommendation comes with a breakdown, not just a score to blindly trust



example output from the app (not to scale)

WHAT I BUILT

A combined query

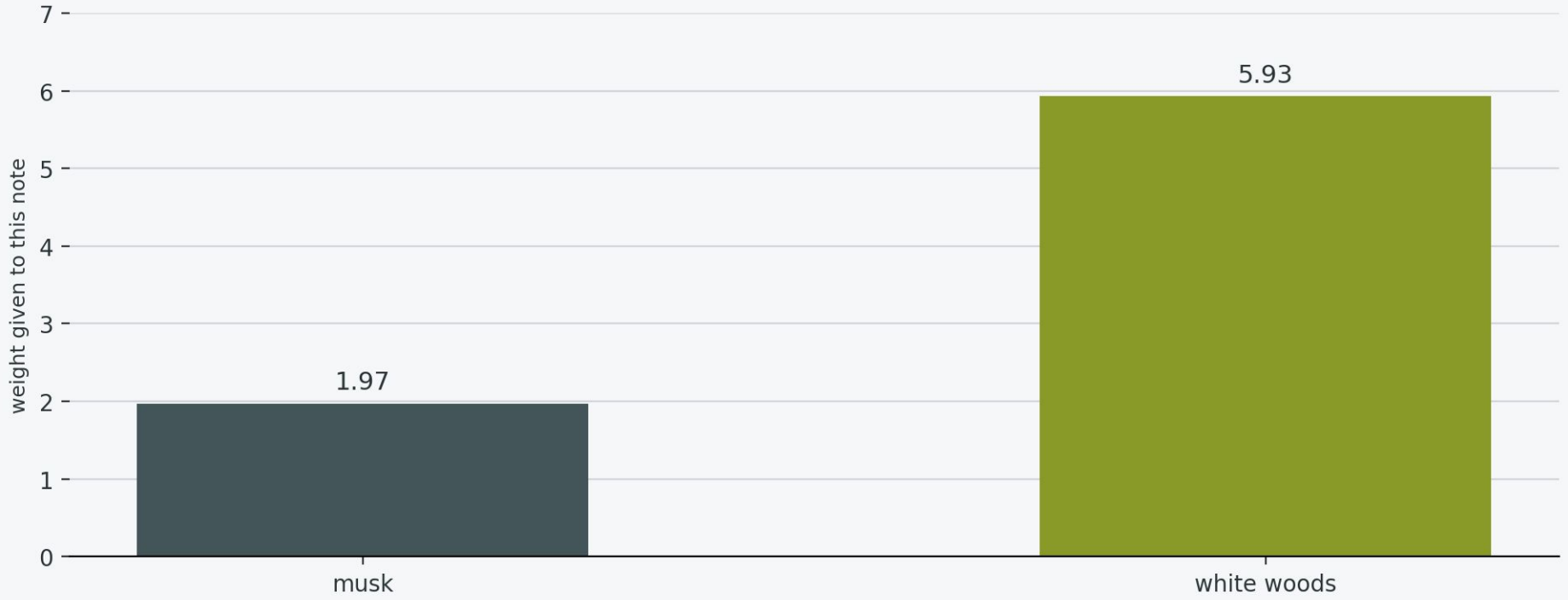
two independent inputs → scored separately → blended into one result



WHAT I BUILT

Rare notes carry more weight

these do the heavy-lifting when it comes to identifying a distinct personal fragrance, so why not prioritise them?



IDF WEIGHT (per note)

```
idf = log(N / (df + 1)) + 1
```

HOW

Quantifying each perfume

NOTES

rare notes
weighted higher

→

ACCORDS

scent families
weighted by strength

→

BLEND

combined
into one score

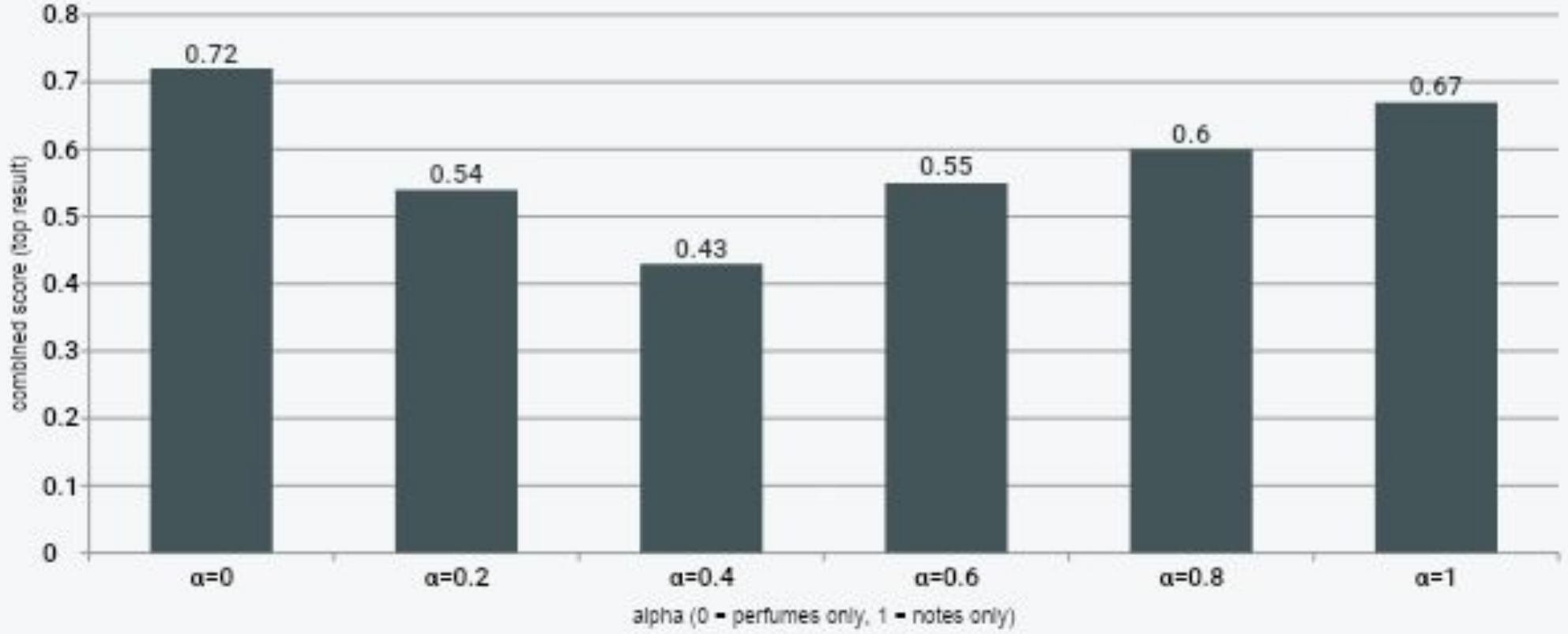
COMBINED PERFUME VECTOR

```
perfume_score = blend_ratio × note_vector + (1 − blend_ratio) × accord_vector
```

HOW

Blended by α

I chose α = 0.5, this chart shows how the combined score changes as α changes from 0 → 1



THE BLEND

```
combined_score = α x notes_score + (1 - α) x perfumes_score
```

HOW

Evaluate as “proof”

except, there's no absolute answer for what smells similar - good thing I know myself and my palate >:)

***negative control*: Paper−**

- ★ a fragrance I know I don’t like must **never** appear in a top recommendations list
  - ☆ a pass/fail sanity check, not a strength measure

note-similarity vs. accord-overlap correlation (*Pearson r*)

- ★ positive correlation with an independently-labelled accord overlap
  - ☆ from Fragrantica's own tags, not my own similarity logic

LIVE DEMO

Accordly app demo

switcheroo

1. **Libre alone**

   top result = just a dupe :/ booooring

2. **+ ambrette**

   diverging individual scores + combined score = completely different results :o

3. **+ white woods**

   1 = notes dominance, 2 = the discovery I crave, 3 = that flipping dupe again D: go awayyyyyy

4. **explained page**

   we pray this works to show my hard work

LIMITS

Ground truths

the circular argument

- ★ tuning alpha against what we already think is similar
  - ☆ the test would just confirm what we already believe then
- ★ same reason precision at top-k was rejected
  - ☆ it would just prove that my top 10 is… my top 10
    (I do deserve a pat on the back tbf)

the dupes & the dupes of dupes & the dupes of dupes of dupes D:

- ★ querying a single (well-known) perfume
  - ☆ the clone(-house) war… they win…
- ★ but the clones are dyslexic :o
  - ☆ impossible to merge every spelling quirk or variation

NEXT

The dataset ratings, vote counts, & have/want data

1

collaborative filtering

right now everything is composition-based, what a fragrance is made of. collaborative filtering would recommend based on behaviour instead, what people with similar taste also liked. a different paradigm, not attempted this capstone.

2

another plausibility check

instead of generating recommendations, that rating and vote data could check them," do top results also share similar ratings or have/want patterns?" another form of supporting evidence, alongside the checks already built.

thank you

<3

